Mini Report

Prompt-based Comparative Evaluation of the Semantic and Pragmatic Competence of Minerva and Gemini

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Introduction

Recent advances in Large Language Models (LLMs) have significantly improved the fluency and grammatical accuracy of artificial language. However, fluent generation does not necessarily demonstrate accurate semantic or pragmatic understanding. Many LLMs still struggle with contextual interpretation, implicit meaning, and ambiguity resolution, especially in multilingual environments. This project comparatively evaluates the semantic and pragmatic competence of Minerva and Gemini LLMs in English context. Minerva AI LLM is the first family of Large Language Models pretrained from scratch in Italian developed by Sapienza NLP in collaboration with Future Artificial Intelligence Research (FAIR) and CINECA[i]. The Minerva models are truly-open (data and model) Italian-English LLMs, with approximately half of the pretraining data composed of Italian text[ii]. The primary aim is to investigate Minerva's ability to interpret meaning beyond literal meaning, given that its primary functional language is Italian. The study focuses specifically on identifying the semantic and pragmatic limitations in Minerva's English outputs.

Research Aim and Question

The main aim of the project is to evaluate and compare the semantic and pragmatic competence of Minerva and Gemini in terms of semantic ambiguity, pragmatic inference, figurative meaning and register-sensitive language. The research question guiding this study is that to what extent Minerva, compared to Gemini, accurately interprets meaning beyond literal meaning in English, particularly in cases involving ambiguity, implication, metaphor and register adaptation.

Dataset and Methodology

The dataset consists of 40 English prompts divided into four linguistic categories including semantic ambiguity, pragmatic, metaphor/figurative meaning and register/style. Each category consists of 10 prompts designed to evaluate different aspects of semantic and pragmatic interpretation. With the aim of a proper conduction of evaluation, every prompt includes the linguistic phenomenon tested, the prompt text, an expected strong answer and a success criterion. Model responses were evaluated using a three-point rubric as follows:

0 = incorrect

1 = partially correct

2 = completely correct

The evaluation specifically focuses on contextual interpretation, ambiguity resolution, pragmatic inference, figurative language understanding and audience-sensitive stylistic adaptation.

Results and Findings

The results show a clear performance difference between Minerva and Gemini across all linguistic categories. Gemini consistently achieved higher average scores and demonstrated significantly stronger contextual and pragmatic reasoning abilities:

Category	Prompt count	Model Minerva Average	Model Gemini Average
Semantic ambiguity	10	0.3	1.9
Pragmatics	10	0.8	2
Metaphor/Figurative meaning	10	1.6	2
Register/Style	10	1.2	1.9

The most significant performance difference emerged in tasks involving semantic ambiguity. Gemini consistently recognized and explained multiple possible interpretations of ambiguous sentences whereas Minerva frequently failed to capture alternative semantic readings. Pragmatics was the second weakest category for Minerva. The model repeatedly struggled with implicit meaning, speech inference and context-dependent interpretation. In many instances, Minerva relied heavily on literal sentence processing and failed to extract implied communicative meaning. In contrast, Gemini consistently identified implied intentions, speaker intent and speech-related inferences. In metaphor and figurative language tasks, both models recognized figurative meaning; however, Gemini produced more semantically nuanced and abstract interpretations. Minerva often reduced metaphorical expressions to simple paraphrases without fully clarifying their conceptual inferences. The results also indicate that Gemini adapted tone, vocabulary and register sensitivity more naturally than Minerva. Overall, the findings suggest that Minerva's English language competence is significantly stronger at the grammatical surface level compared to its deeper semantic and pragmatic levels of interpretation.

Limitations

The project has several limitations. First, the dataset is relatively small, containing only 40 prompts. Second, the evaluation process relies on manual scoring; not on automatic evaluation metrics, which introduces subjectivity. Lastly, model outputs may vary between sessions due to the probabilistic nature of LLM generation.

Conclusion

This project demonstrates that, although Minerva can generate fluent and grammatically coherent English responses, the model still exhibits important limitations in semantic and pragmatic interpretation when compared to Gemini. This suggests that Minerva’s semantic and pragmatic competence in English has not yet reached the same level of development as its primary Italian-language performance.

Overall, the project highlights several areas in which Minerva could be improved. Future development should focus on strengthening contextual reasoning, ambiguity resolution, pragmatic inference and register/style adaptability in English-language outputs. Improving these aspects could significantly enhance Minerva’s ability to interpret meaning beyond literal sentence structure and produce responses that are semantically and pragmatically more accurate.

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[i] https://minerva-ai.org/

[ii] https://minerva-ai.org/